



NO BOUNDARIES: Nair
with a launch model



SANJAY RAMCHANDRAN

INDIAN SPACE RESEARCH ORGANISATION, BANGALORE

FLYING HIGH

He dreamt big but even Vikram Sarabhai would have been surprised at how vaulting his ambition has become, had he seen the satellite launch vehicle (SLV) in April 2008 that lobbed 10 satellites, including eight foreign ones, in one go. It was his baby—the Indian Space Research Organisation (ISRO), headquartered in Bangalore that had taken the giant step. And now ISRO is taking an even bigger step—a maiden journey to the moon with Chandrayaan I which will explore it using a robotic spacecraft, weighing 1,300 kg and carrying 11 experiments including six from NASA and the European Space Agency. Groundwork in the form of a huge antenna, measuring 105 ft, an associated electronic system and a centre to archive scientific data from Chandrayaan has already been done at Bylalu village near Bangalore. The made-in-India antenna is sensitive enough to track spacecraft crores of kilometres away—even near Mars. Such accomplishments have made India a world leader in the civilian remote sensing field with eight satellites providing images to not only India

“Our thrust is to harness technology for down-to-earth applications.”

G. MADHAVAN NAIR, CHAIRMAN, ISRO

but the global market as well. One of these can spot features as small as three ft from a height of 600 km. Indian space scientists, working on shoestring budgets, have created a great impact on telecom, television broadcast and weather watch, particularly in reaching out to the rural masses in the fields of education and healthcare. Now ISRO is fast tapping the lucrative market of launching payloads of other nations, among them spy satellites for Israel and a recent payload for Indonesia. Currently, ISRO is leveraging its expertise in cryogenic technology to design hydrogen fuel cells to store hydrogen, and has teamed with Tata Motors to develop a prototype of a hydrogen passenger car to be launched in the Indian market later this year. Indeed as ISRO Chairman G. Madhavan Nair, who has established a space science university in Kerala, puts it, “Sky is not the limit here.”

by Stephen David

1962 was the year when church property was transferred to the Thumba station in Kerala for rocket launches.